

Question 31 / Vraag 31

In general, this question was not answered well. Many students erroneously referred to money being transferred by the International Monetary Fund (IMF) and that this then leads to an increase in spending (and GDP) in the economy. Some students also erroneously said that unpaid household work will be paid for and that this results in an increase in GDP. This is not correct. If you read the question carefully, you will see that a research paper by the IMF found that policies targeted at redistributing unpaid household work across men and women could increase GDP by 4 percent. This unpaid household work refers to work such as childcare, cooking and cleaning that women tend to do in their own households (but that they are not paid for). When these tasks are redistributed across men and women, it means that men perform a bigger share of these unpaid household work, which then frees up time for women to become more involved in the labour market and take up paid work. When women take up more paid work, their income increases, and GDP of the country also increases.

/ Oor die algemeen is hierdie vraag nie goed beantwoord nie. Baie studente het verkeerdelik verwys na geld wat oorgeplaas word deur die Internasionale Monetêre Fonds (IMF) en dat dit dan lei tot 'n toename in besteding (en BBP) in die ekonomie. Sommige studente het ook verkeerdelik genoem dat die onbetaalde huishoudelike werk betaling sal ontvang en dat dit sal lei tot 'n toename in BBP. Dit is nie korrek nie. Indien jy die vraag versigtig lees, sal jy sien dat die navorsingsartikel deur die IMF bevind het dat beleid wat daarop gemik is om onbetaalde huishoudelike werk tussen mans en vroue te herverdeel, BBP met 4 persent kan verhoog. Hierdie onbetaalde huishoudelike werk verwys na take soos versorging van kinders, kosmaak en skoonmaak wat vroue geneig is om in hulle eie huishoudings te verrig (maar waarvoor hulle nie betaling ontvang nie). Wanneer hierdie take herverdeel word tussen mans en vroue, beteken dit dat mans 'n groter deel van hierdie onbetaalde huishoudelike werk verrig, wat dan tyd losmaak vir vroue om meer betrokke te raak by die arbeidsmark en meer betaalde werk te doen. Wanneer vroue meer betaalde werk verrig, styg hulle inkomste, en gevolglik neem die BBP van die land ook toe.

Q 32

32.1. About half of the students got this question correct. It is clear that there was confusion between Pareto dominance and technological dominance – put simply, when determining whether an allocation Pareto-dominates, you would typically look for the allocations with the larger pay-offs; whereas where you are determining technological dominance, you are looking for the most efficient technologies – thus, those technologies with the lowest inputs.

32.2. Well answered. Many students stopped at the first step: $300\,000 = 900L + 3000G$; But the equation must be simplified to $G = 100 - 0.3L$

32.3. The slope of the isocost line can be represented by the MRT (1/2 mark), but the line itself is the relative cost (in this case, the relative cost of workers). Many students confused MRT with MRS (the (-)slope of the indifference curve).

Q33

In general, most students did not get full marks for this question. Please make sure that you know your formulas very well, but also the meaning of terminology such as marginal product (MP) and average product (AP).

33.1 Marginal product (MP) refers to the additional unit(s) that are produced as you increase the amount of labour in the production process.

Using the mid-point method, we can calculate marginal product.

$$\text{Marginal product (MP)} = (\text{Change in } Q)/(\text{Change in } L) = (750-300)/(45-15) = 15$$

$$\text{Therefore, } MP(30) = 15$$

$$33.2 \text{ Average product (AP)} = \text{Total output (Q)}/\text{Labour (L)} = 600/30 = 20$$

$$\text{Therefore, } AP(30) = 20$$

Oor die algemeen het die meeste studente nie volpunte vir hierdie vraag gekry nie. Maak asseblief seker dat jy jou formules baie goed ken, maar dat jy ook die betekenis van terminologie soos marginale produk (MP) en gemiddelde produk (AP) ken.

33.1 Marginale produk (MP) verwys na die ekstra eenheid (of eenhede) wat geproduseer word soos wat jy die hoeveelheid arbeid in die produksieproses verhoog.

Deur van die middelpunt-metode gebruik te maak, kan ons die marginale produk bereken.

$$\text{Marginale produk (MP)} = (\text{Verandering in } Q)/\text{Verandering in } L = (750-300)/(45-15) = 15$$

$$\text{Dit beteken dat } MP(30) = 15$$

$$33.2 \text{ Gemiddelde produk (AP)} = \text{Totale uitset (Q)}/\text{Arbeid (L)} = 600/30 = 20$$

$$\text{Dit beteken dat, } AP(30) = 20$$

Q34

34.1. & 34.2. Most students did well. Those who got it incorrect must have simply calculated how many glasses of wine and camembert cheese Simon could afford given his income.

34.3. Most students struggled with this question. By using the definition $MRS=MRT$, students could have determined the slope of the budget line intersecting C – which was $10/3 = 3.3333$

In general, students did not do well in this question.

35.1 To determine the Nash equilibrium in this game, you need to ask yourself the following questions:

If Player 1 chooses to play strategy A, what is the best that Player 2 can do? It is to play strategy B and get a payoff of 4.

If Player 1 chooses to play strategy B, what is the best that Player 2 can do? It is to play strategy A and get a payoff of -4.

If Player 2 chooses to play strategy A, what is the best that Player 1 can do? It is to play strategy B and get a payoff of 4.

If Player 2 chooses to play strategy B, what is the best that Player 1 can do? It is to play strategy A and get a payoff of -4.

There are therefore two Nash equilibriums, i.e., (B,A) and (A,B). These equilibriums can also be written as (4,-4) and (-4,4).

35.2 No, since there are two Nash equilibriums, there is no dominant strategy. The game in this question is an example of a chicken game.

Oor die algemeen het studente nie goed gevaar in hierdie vraag nie.

35.1 Om die Nash-ewewig in hierdie spel te bepaal, moet jy jouself die volgende vrae afvra:

Indien Speler 1 strategie A speel, wat is die beste wat Speler 2 kan doen? Dit is om strategie B te speel en 'n uitbetaling van 4 te kry.

Indien Speler 1 strategie B speel, wat is die beste wat Speler 2 kan doen? Dit is om strategie A te speel en 'n uitbetaling van -4 te kry.

Indien Speler 2 strategie A speel, wat is die beste wat Speler 1 kan doen? Dit is om strategie B te speel en 'n uitbetaling van 4 te kry.

Indien Speler 2 strategie B speel, wat is die beste wat Speler 1 kan doen? Dit is om strategie A te speel en 'n uitbetaling van -4 te kry.

Daar is dus twee Nash-ewewigte, nl. (B,A) en (A,B). Hierdie twee ewewigte kan ook geskryf word as (4,-4) en (-4,4).

35.2 Nee, aangesien daar twee Nash-ewewigte is, is daar geen dominante strategie nie. Die spel in hierdie vraag is 'n voorbeeld van 'n hoenderspel.

Q36

36.1 Most students got at least 2 out of 3 for this question. Students got a mark for stating that it is Pareto Inefficient; Another mark for arguing that there is a Pareto improvement possible. For the full 3 marks, students must have demonstrated how a Pareto improvement could take place by the individuals swapping their allocated flavours among each other.

36.2 About a third of students got this correct. There seems to be confusion on what Pareto-efficient means: i.e. all resources are being spent AND no Pareto improvements are possible from a specific allocation. If Johan steals all of the ice-cream, it is in fact Pareto efficient, because no Pareto-improvement is possible without making Johan worse off. Some students correctly stated that this is unfair, but it is still Pareto-efficient.

Q37

37.1. Most students got this correct. Those who did not find the right answer were confused about what the slopes of the Isocost curves represent – the steeper the isocost slope in this model, the lower the cost of effort. In other words, a small increase in the Wage (cost of labour) leads to a large change increase in associated effort; therefore, the steeper the slope, the lower the cost of effort.

37.2. About 2/3 students got this correct. The optimal wage is represented by the point where the Isocost line is tangential to the best response curve (in this case, point A). BUT – the question required students to write down the wage at this point, which can be read from the figure.

37.3. Most students correctly stated point A and also correctly stated that this is because it lies on Oceanus' best response curve; but also then included point C as a feasible option. Feasibility implies not only what is possible, but what is optimal for the company. While C is a possible point, it being on the highest cost of effort isocost curve is not feasible.

Q38

38.1. Students fared well in this question. Some students calculated the surplus in this market in terms of the price – this is incorrect. Surplus refers to the excess quantity being produced.

38.2. Most students got this correct. Some students stated that a price floor has no effect on consumer surplus, which would be true if the price floor is not binding. However, this is not the case in this question, where the binding price floor reduces consumer surplus.

38.3. Students did fairly well here. Mostly students struggle to identify where the DWL. Sometimes, students would indicate the DWL as a line, or as the area below the demand or supply curves, which is incorrect. It forms a triangle between the demand and supply curves.

Many students also failed to include the necessary labels – always show the axis intercepts, label the demand and supply curve, the axes, and colour in the area of the DWL.

Q39

Around half of students got full marks for this question. Students needed to recognise that cigarettes are a good with inelastic demand (because it is incredibly addictive). Therefore, price increases due to taxes would not lead to a proportionally large decrease in demand. Therefore, the minister can exploit this fact and increase tax revenue while increasing taxes on this inelastic product.

Q40

40.1. It is clear that most students have not mastered price elasticity. And to a certain extent it is very clear that most students do not know how to calculate percentage increases/decreases. There are multiple formulas for price elasticity and in this question students could have made use of any of these formulas. E.g. $\frac{\Delta Q}{Q} / \frac{\Delta P}{P}$ or $\frac{\Delta Q}{\Delta P} \times \frac{P}{Q}$.

Some students that correctly stated the formula $\frac{\Delta Q}{Q} / \frac{\Delta P}{P}$ then went on to calculate the changes in percentages wrongly – this is a simple, but incredibly useful calculation that all BComm students must master.

40.2 Most students correctly stated that the shop owner should lower the price, but the explanation was wrong mostly. Many students stated that decreasing the price will lead to an increase in demand – which is simply the law of supply and demand. Price elasticity asks whether you should change the price based on its effect on your REVENUE and NOT the DEMAND. Many students also stated that his profits would increase if he decreased the price of Pepsi, which is not entirely correct – it is more fitting to state that revenue will increase.

Q41

41.1. In this question, less than 5% of the class could achieve the full 4 marks. Most students could recognize that a shortage would occur in the market (1 mark), but failed to discuss the other welfare effects (decrease in producer surplus, increase in consumer surplus, increase in DWL due to unexploited gains from trade).

41.2. Most students got this correct – you simply had to recognise that the price ceiling is non-binding (has no effect) because the market set the price below the ceiling anyway.

Q42

Very few students got all four answers correct, but mostly got 2/4.

42.1.1. Students had to recognise the price where $S = D$; which is C

42.1.2. Similarly, the quantity where $S = D$; which is F

42.2. Students struggle overall to recognise the DWL; not just in this question. The answer was NKM.

42.3. Most students correctly stated KN.

